Aman Sharma

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Education

Indian Institute of Technology Delhi — New Delhi, India
Ph.D. Civil Engineering (Transportation) — 2020-2026
Thesis title: Reliable electric bus system design considering energy supply uncertainties

NIT Hamirpur — Hamirpur, India
M.Tech. Transportation Systems Engineering — 2018-2020
Thesis title: Optimal generation of fast transit corridors in a city

G.B. Pant Engineering College — Pauri Garhwal, India
B.Tech. Civil Engineering — 2013-2017

Research Interests

Addressing challenges in real-world decison making due to uncertainties and disruptions. Special interests include planning and optimizing large-scale transportation systems and their operations, integrating clean energy, sequential decision making, and addressing uncertainties for resilient and reliable operations using operations research techniques.

Journal Publications

- Sharma, A., & Nezamuddin, N. (2026). Electric bus system design reliable against stochastic power grid load shedding and uncertain charging efficiency. *Reliability Engineering & System Safety, 273*, 112347. https://doi.org/10.1016/j.ress.2026.112347
- Sharma, A., & Nezamuddin, N. (2026). Electric bus system design considering power grid disruptions. *Transport Policy, 179*, 104015. https://doi.org/10.1016/j.tranpol.2026.104015
- Sharma, A., Gairola, P., & Nezamuddin, N. (2026). Resilient electric bus system design considering charger failure. *Operational Research, 26*(2), 37. https://doi.org/10.1007/s12351-026-01024-5
- Nasar, A., Sharma, A., & Nezamuddin, N. (2025). A comparative analysis of in-motion and overnight charging infrastructure design for e-buses. *Energy for Sustainable Development, 88*, 101784. https://doi.org/10.1016/j.esd.2025.101784
- Sharma, A., & Parti, R. (2021). Optimal generation of fast transit corridors in a city. *Current Science, 120*(9), 1500-1506. https://doi.org/10.18520/cs/v120/i9/1500-1506

Conferences

- Sharma, A. & Nezamuddin, N. Electric bus system design considering power grid disruptions. *Conference on Advanced Systems in Public Transport (CASPT)* 2025, Kyoto, Japan.
- Sharma, A., Das, S., Nasar, A., Gairola, P., Gandhi, S., & Nezamuddin, N. (2025). Evaluation of electric bus operations in hilly terrain—a case study of Leh. In: Sahu, P.K., Agarwal, A., Kathuria, A., Velaga, N.R. (eds) *Proceedings of the 7th International Conference of Transportation Research Group of India (CTRG 2023), Volume 3.* CTRG 2023. Lecture Notes in Civil Engineering, vol 422. Springer, Singapore. https://doi.org/10.1007/978-981-97-9943-5_2
- Sharma, A., Gairola, P., & Nezamuddin, N. Delay minimization for uncertain electric bus operations due to charging infrastructure breakdown. *23rd Conference of the International Federation of Operational Research Societies (IFORS)*, 2023, Santiago, Chile.

- Sharma, A., Gairola, P., & Nezamuddin, N. Optimal electric bus system design considering uncertainty in charging infrastructure mechanism. *102nd Annual Meeting of the Transportation Research Board (TRB)*, 2023, Washington, DC, USA.
- Sharma, A., Gairola, P., & Nezamuddin, N. Electric bus system charging infrastructure design with fleet determination considering dwell times. *6th International Conference on Intelligent Traffic and Transportation (ICITT)*, 2022, Paris, France.

Project Experience

Research Assistant and grant writing with Prof. Nezamuddin at IIT Delhi in the following projects:

1. Bus Electrification and Scheduling Framework Under Operational Perturbations 2024-2027
   Sponsoring Agency: Anusandhan National Research Foundation (ANRF), Government of India.
2. Catenary Based In-Motion Charging Mechanism for Electric Buses 2023-2024
   Sponsoring Agency: Bureau of Energy Efficiency (BEE), Ministry of Power, Government of India.
3. Electric Bus Operation and Performance Data Collection and Analysis 2022
   Sponsoring Agency: SG Architects, New Delhi.

Teaching Experience

Teaching assistant of Prof. Nezamuddin at IIT Delhi for the following courses: Intelligent transportation systems; Transportation logistics; Transportation economics.
Mentored several B.Tech. and M.Tech. projects during this duration.

Certification/Courses

- Summer School on Large Scale Optimization organized by Indian Institute of Technology Delhi, June 2026
- Summer School on Large Scale Optimization organized by Indian Institute of Technology Kanpur, April 2023
- E-bus System Planning and Optimization from pManifold EV Academy, August 2021
- Linear and Network Optimization organized by Indian Institute of Science Bengaluru, September 2020
- Credit courses on Data analysis for transportation engineering; Urban and regional transportation; Transportation network optimization at Indian Institute of Technology (IIT) Delhi, India

Skills & Interests

Technical: GAMS; GeoPandas; GIS; Gurobi; LaTeX; Linux; Python; R; SQL.
Language: English (fluent); Hindi (native).

References

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Dr Pranav Gairola
CRF (Centre Research Fellow),
Brij Disa centre for Data Science and Artificial Intelligence,
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